

LEARNING FROM DEAD

Postmortem Examination of Asian Elephant

Comprehensive Study Notes with Slide Images

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Wildlife Health - An important and still growing new discipline, yet often ignored. Though an imp component for conservation of wildlife.

Wildlife Health Management:
Wildlife Health Management is a branch of Veterinary Science which deals with veterinary interventions in wild fauna i.e health monitoring, disease management, scientific immobilization, and habitat manipulation in order to achieve conservation strategies to maintain healthy population or conservation of a particular species.

Shrivastav, 2010

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Slide 1 — Title slide: NDVSU Jabalpur and credentials of Prof. Dr. A B Shrivastava

1. Wildlife Health Management — Foundations

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Post-mortem examination –
Examination of external surfaces and internal organs for presence of pathological or abnormality in a carcass to ascertain the cause of death.
Roman physician Antistius - earliest autopsies in 44 BC of Julius Caesar and documented 23 wounds.
In 1410, Catholic Church ordered an autopsy on Pope Alexander V to determine poisoning.
Surgeon Bulkeley – 1st PM in India in 1680.
Sushruta in 2500 years BC - The Sushruta Samhita systematic method for the dissection of the human cadaver.



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Slide 2 — Wildlife Health Management: Multidisciplinary Approach diagram

1.1 Definition

Wildlife Health Management is a branch of Veterinary Science dealing with veterinary interventions in wild fauna — health monitoring, disease management, scientific immobilisation, and habitat manipulation to achieve conservation strategies for healthy populations. (Shrivastav, 2010)

1.2 Multidisciplinary Disciplines

Discipline	Relevance to Elephant Health
Wildlife Pathology	Core of mortality investigation; gross and histopathological disease ID
Wildlife Anatomy	Normal vs pathological anatomy at necropsy; species-specific features
Wildlife Clinics & Ethology	Pre-mortem clinical signs; normal behavioural baselines
Wildlife Parasitology	Internal/external parasite ID; zoonotic risk
Genetic Mgt of Small Populations	Conservation genetics; inbreeding; PVA
Animal Biotechnology	PCR diagnostics; DNA forensics; EEHV vaccine development
Wildlife Microbiology	Bacterial/viral culture; AMR profiling; zoonosis
Wildlife Nutrition	Nutritional deficiency diseases; body condition scoring

KEY: Wildlife Health is often ignored — yet every elephant death is data that can protect the living population. Systematic necropsy is not optional — it is a conservation imperative. "Carcass never tells lie." — Dr Satpati

2. History of Postmortem Examination

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INTRODUCTION

- Animals & birds - Necropsy
- Human - Autopsy
- Postmortem examination should be conducted as per the standard protocol and with adequate facilities.
- Carcass of wild animal is seldom found in a suitable condition for PM examination
- Who can perform the postmortem of wild animals ?
Experienced Wildlife Veterinarian

Postmortem examination is an art of disease diagnosis. Scientific and systematic is an essential requirement for recording detailed gross pathologica; and also postmortem changes in the carcass. Many times PM alone and gross pathological lesions, and their correlation with history may not be sufficient to make diagnosis.

For confirmative diagnosis laboratory support is essential, for which most suitable biological samples in a appropriate preservative should be collected. Badly putrified sample or fresh sample in a unsuitable preservative will not be useful for lab examination and confirmative diagnosis.

Staff	required	2-3 vets	4 supporting staff for PM
		1 Photographer	1 for sample collection

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Slide 3 — Introduction: Necropsy (animals) vs Autopsy (humans); standard protocol requirements

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• **Postmortem examination:**
As per the standard protocol and with adequate facilities. Carcass of wild animal is seldom found in a suitable condition for PM examination

• Who can perform the postmortem of wild animals
Experienced Wildlife Veterinarian / Veterinary Pathologist having exposure of wildlife

“ Doctors who perform necropsies or who regularly witness postmortem examinations, learnt to solve their doubts in comparison to those who are not dealing necropsy are floating in the air.
Dr A B Shrivastav, 2014

• Carcass never tells lie Dr Satpati

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Slide 4 — Professional quotes on necropsy importance; "Carcass never tells lie" — Dr Satpati

2.1 Historical Milestones	
Era / Person	Contribution
Sushruta, 2500 BC	Sushruta Samhita: systematic human cadaver dissection in ancient India
Roman physician Antistius, 44 BC	Earliest documented autopsy — Julius Caesar; 23 wounds identified; first forensic pathology
Catholic Church, 1410	Autopsy on Pope Alexander V to determine poisoning; medicolegal precedent
Surgeon Bulkley, 1680	First documented postmortem in India
Modern Era	Standardised protocols; histopathology; molecular diagnostics; digital documentation

2.2 Key Principles	
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- NECROPSY (animals & birds) vs AUTOPSY (humans) — same process, different terminology
- PM is an ART of disease diagnosis — scientific and systematic approach is essential
- Gross findings + history alone may be insufficient — laboratory confirmation required
- Badly putrefied samples or fresh samples in wrong preservatives are useless
- Staff: 2–3 vets + 4 supporting + 1 photographer + 1 sample collection personnel

3. Necropsy Objectives & Pre-Necropsy Evaluation

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Necropsy Objectives

- To know all possible lesions of disease/abnormality.
- To correlate lesions with clinical and laboratory findings.
- To understand the sequence of disease events.
- To prevent or control the disease in healthy population

Pre-necropsy Evaluation

- Examine the necropsy letter for request for the following:
 - i. Specific lesions in different organs or systems desired .
 - ii. Whether it is a cosmetic or routine necropsy.
 - iii. Special requests (i.e. sample collection for specific investigations / cultures / photos, etc).
- Identification of the carcass to ensure that the correct carcass is being necropsied.
- Clinical and circumstantial history .
 - P M examinations of Sch I animals should be performed systematically, sincerely and seriously , not a routine chore.

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Slide 5 — Necropsy Objectives; Pre-necropsy Evaluation checklist

3.1 Necropsy Objectives

- To know ALL possible lesions of disease or abnormality
- To CORRELATE lesions with clinical and laboratory findings
- To understand the SEQUENCE of disease events
- To PREVENT or CONTROL disease in the healthy living population

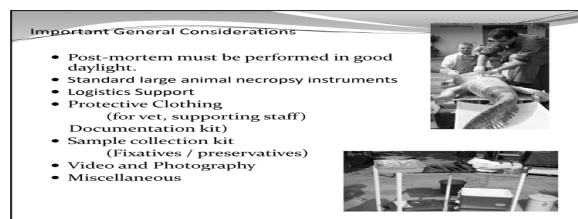
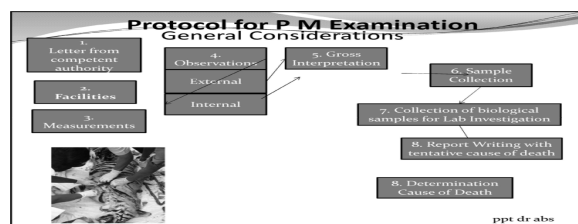
3.2 Pre-Necropsy Evaluation Checklist

Step	Action	Key Points
1	Examine necropsy request letter	Specific lesions desired? Cosmetic vs routine? Special samples needed?
2	Identify carcass	Species, sex, microchip, GPS collar, physical markings — ensure correct carcass
3	Collect clinical history	Cause of death (as reported), clinical signs, treatment given, PMI
4	Assess PM interval	Critical for autolysis grade and sample quality assessment
5	Legal authorisation	DFO/CCF letter mandatory for Schedule I species

LEGAL: PM examinations of Schedule I animals (including Asian elephant) must be performed systematically, sincerely and seriously — NOT a routine chore. Under WPA 1972, negligent necropsy is a criminal offence.

4. Protocol for PM Examination & Equipment

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Slide 6 — PM Protocol general considerations flowchart; field necropsy photographs

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Equipment required for postmortem examination:

S.no.	Item	Minimum no.
1.	Copy of Necropsy protocol	1
2.	Hand held GPS	1
3.	Digital camera	1
4.	Digital video camera	1
5.	Clip board	2
6.	Measuring tape	1
7.	Plastic ruler 15cm. (clear markings)	2
8.	Vernier callipers	1
9.	Portable weighing machine (250 kg)	1
10.	Necropsy format	5
11.	Ballpen	2
12.	Pencil rubber	1
13.	Metal detector	1

Equipment required for postmortem examination best SS.quality :

Sn.no	Equipment	Minimum no.
1.	Sharp high quality necropsy knife ss	4
2.	Skinning knife curved ss	2
3.	Autopsy knife curved ss	2
4.	Slicing knife 22 mm blade ss	1
5.	Knife sharpener stone or steel	1
6.	Bard Parker Handle and blades (No. 20,22,26 (BP)	5 each
7.	Small plain forceps	2
8.	Artery Forceps straight 203mm ss	2
9.	Dissecting Forceps 200mm long ss	2
10.	Large Mayo dissecting scissors fine point 215mm	2
11.	Bone Cutter Compound action 266mm	1
12.	Hack saw or bone saw blade 254mm	1
13.	Chisels 8 1/2" X 1 1/4" (22.00cm with blade 3.00 cm)	1
14.	Hammer Wrench end 200mm	1
15.	Portable autopsy saw (Electric/battery)	1
16.	Axe (Roofing axe SS) 60mm blade	1
17.	Spirit lamp SS or gas burner	1
18.	Measuring tape SS	1

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Slide 7 — Equipment lists: documentation items (13) and surgical instruments (18)

4.1 General Considerations

- Perform in GOOD DAYLIGHT — artificial light misses subtle colour changes (icterus, cyanosis)
- Standard LARGE ANIMAL necropsy instruments — domestic instruments are inadequate
- Protective clothing: waterproof overalls, double gloves, goggles, face mask, gumboots — for ALL personnel
- Sample collection kit: 10% buffered formalin, 70% ethanol, sterile swabs, EDTA tubes, biohazard bags
- METAL DETECTOR: mandatory before any incision — check for bullets/metallic foreign bodies

4.2 PM Protocol Steps

Step	Action
1	Letter from competent authority (DFO/CCF)

2	Assess facilities — field vs laboratory; natural light; water access
3	Measurements: weight, length, girth, tusk, foot circumference
4	External observations — before any incision
5	Internal observations — cavity by cavity, system by system
6	Gross interpretation — correlate all findings; differential diagnosis
7	Sample collection — tissues for histopathology, culture, PCR, toxicology
8a	Report writing — tentative cause of death + all gross findings
8b	Final cause of death — after laboratory results

4.3 Equipment — Documentation (13 Items)

S.No.	Item	No.
1	Copy of Necropsy Protocol	1
2	Hand-held GPS	1
3	Digital camera	1
4	Digital video camera	1
5	Clip board	2
6	Measuring tape	1
7	Plastic ruler 15 cm (scale bar for photos)	2
8	Vernier callipers	1
9	Portable weighing machine 250 kg	1
10	Necropsy format (forms)	5
11	Ballpen	2
12	Pencil rubber	1
13	Metal detector	1

4.4 Equipment — Surgical Instruments SS Quality (18 Items)

S.No.	Instrument	No.
1	Sharp high-quality necropsy knife SS	4
2	Skinning knife curved SS	2
3	Autopsy knife curved SS	2
4	Slicing knife 22 mm blade SS	1
5	Knife sharpener stone or steel	1
6	Bard Parker Handle & blades No.20,22,26	5 each
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9	Dissecting Forceps 200 mm long SS	2
10	Large Mayo scissors 215 mm fine point	2
11	Bone Cutter Compound action 266 mm	1
12	Hack saw/bone saw blade 254 mm	1
13	Chisels 8½" × 1¼" (22 cm, blade 3 cm)	1
14	Hammer Wrench end 200 mm	1
15	Portable autopsy saw (Electric/battery)	1
16	Axe (Roofing axe SS) 60 mm blade	1

17	Spirit lamp SS or gas burner	1
18	Measuring tape SS	1

5. External Examination

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1. History of animal.

- Information on – probable cause of death, clinical signs and treatment given, if known.
- Information on the circumstances of death.

External Examination

Body Condition

Rigor Mortis

External Injuries

Characterization of injuries

Documentation

Any other observation



Postmortem Examination

2. External Examination

- Identify sex – Examine male/female external organs . Also see whether an animal is pregnant, lactating or dry.
- Nutritional status of the carcass – Prominent ribs, depression in lumbar, buccal region. distinct temporal fossa, loose skin - suggest poor health.
- If possible search for dislocations and fractures,
- Presence of algor mortis, rigor mortis and livor mortis autolytic changes.
- Examine visible m m

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Slide 8 — External examination: history, body condition, rigor mortis, injuries; carcass photograph

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v. Skin for external parasites, cutaneous filariasis and lesions of pox, oedematous swellings, emphysematous crepitation, injuries, burns, warts and the visible mucous membranes for pallor, congestion, cyanosis, and inflammatory conditions.

vi. Examine injury – bullets, intra-species fight, and mauling by large carnivores (in calves).
For captive elephants- injuries inflicted by mahouts (ankush).

vii. Examine dung bolus of the rectum - indication about digestive status /dentition/ hydration status

viii. Natural orifices- discharges and the oral cavity for lesions foot-and-mouth disease, EEHV, anthrax etc.

ix. Examine temporal glands for musth

x. Foot pad and nails for injuries, healthy appearance.

Detailed examination:

- Cadaveric lividity (hypostatic congestion):
Common - large animal like elephant.
Irregular patches may observed - s/c tissues of the side upon which the animal is lying for long time.
- Rigor mortis:
Stiffing of the carcass develops soon after death.
Influenced by season, nutritional status / disease.
Weak and emaciated diseased /septicaemia – less marked
Usually - appears within 1-4 hours after death and may lasts for 16-24 hrs and, at times, may seen up to 48 hrs depending on factors.
- Autolytic changes :
Decomposition begins soon after death also depends upon the atmospheric moisture, temperature and PM interval.

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Slide 9 — Skin examination, orifices, temporal glands, dung bolus, post-mortem changes

5.1 Initial Survey

- **HISTORY:** probable cause of death, clinical signs, treatment given, circumstances, PMI
- **BODY CONDITION:** prominent ribs, lumbar/buccal depression, distinct temporal fossa, loose skin = poor health
- **RIGOR MORTIS:** appears 1–4 hrs after death; lasts 16–24 hrs (up to 48 hrs); less marked in emaciated/septicaemic animals
- **LIVOR MORTIS (Cadaveric Lividity):** irregular subcutaneous patches on lying side — **NORMAL**, not haemorrhage
- **DOCUMENTATION:** photograph EVERY finding with scale bar **BEFORE** any manipulation

5.2 Detailed 10-Point External Protocol

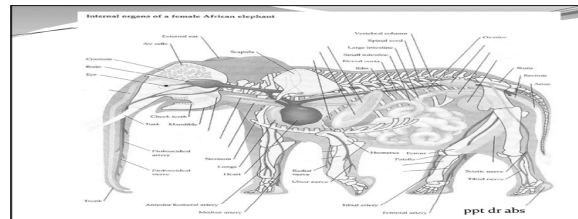
Point	Examination	Key Findings to Document
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i	Sex — male/female external organs; pregnant/lactating?	Gravid uterus; mammary engorgement; scrotal absence (normal in males)
ii	Nutritional status	Prominent ribs; temporal/lumbar/buccal depression; loose skin → emaciation
iii	Fractures & dislocations	Limbs, ribs, skull — HEC injury, train impact, fall
iv	Post-mortem changes (algor/rigor/livor/autolysis)	Degree of PM changes → estimate PMI
v	Skin: parasites, filariasis, pox, oedema, emphysema, burns, warts, mucous membranes	Crepitation (gas gangrene/Clostridium); cherry-red (CO/HCN); cyanosis
vi	Injury: bullets, intra-species, carnivore mauling (calves), ankush (captive)	Entry vs exit wounds; mauling patterns; ankush punctures
vii	Dung bolus of rectum	Coarse undigested material = dental exhaustion; dry = dehydrated; liquid = diarrhoea
viii	Natural orifices — discharges; oral cavity (FMD, EEHV, anthrax)	EEHV haemorrhagic oral lesions in calves; FMD vesicles; anthrax black discharge
ix	Temporal glands — musth status	Active secretion in musth; swelling/discharge outside musth = pathological
x	Foot pads and nails — injuries, fissures, abscess	Captive: foot pad disease, osteomyelitis; wild: normal even wear

6. Dentition & Age Determination

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Dentition		
Elephants develop six sets of molars, and these can be used for age determination. The normal molar pattern of eruption is as follows:		
M1	1 year	5 laminae
M2	2 years	7 laminae
M3	6 years	10 laminae
M4	15 years	10 laminae
M5	28 years	12 laminae
M6	47 years	13 laminae
The permanent tusks protrude beyond the lips at about 30 months and grow throughout life.		



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Slide 10 — Molar eruption table; lamellae count for age estimation; internal organ anatomy diagram

6.1 Horizontal Molar Progression — Unique to Elephants

Six successive molar sets (M1–M6) develop sequentially at the BACK of the jaw and move FORWARD. When M6 (the final molar) wears out, the animal can no longer chew → starvation → death. This sets the biological maximum lifespan at ~60–70 years.

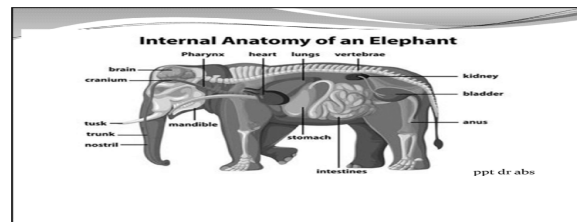
6.2 Molar Eruption Table

Molar	Eruption Age	Lamellae	Active Use Period	Notes
M1	~4 months	5	0–1 yr	Deciduous; shed rapidly
M2	~1–2 yrs	7	1–4 yrs	Milk tooth
M3	~6 yrs	10	4–15 yrs	First large molar
M4	~15 yrs	10 WIDER	15–28 yrs	Same count as M3 but WIDER lamellae
M5	~28 yrs	12	28–47 yrs	Longest in use; prime adulthood
M6	~47 yrs	13	47 yrs–death	Final molar; never replaced

- Permanent tusks protrude beyond lips at ~30 months; grow throughout life
- Count enamel plates (lamellae) with probe AFTER cleaning — do not estimate visually
- Distinguish M3 (10 narrow plates) vs M4 (10 WIDE plates) — different age ranges

7. Internal Anatomy & Skinning

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Skinning of elephant carcass

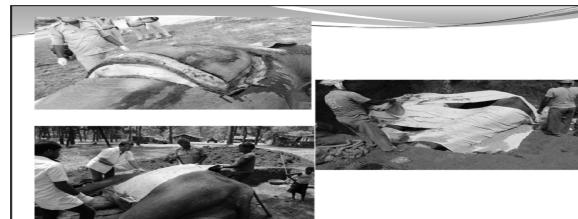
In healthy animal, the skin is thick but flexible and easily moved over the underlying tissues. It is essential to remove skin of one side from the entire lateral thoracic and up to lower abdominal and other suspected regions.

Examine the subcutaneous tissues and muscles for abscesses, wounds, haemorrhages and other pathological conditions.
Examine superficial lymph nodes

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Slide 11 — Labelled internal anatomy diagram of elephant (cross-sectional)

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Digestive system: From cavity - pharynx, oesophagus, simple stomach, small and large **intestines**, caecum, rectum and anus.

Additional organs - molar teeth, tongue, salivary glands, liver and pancreas.

Elephant - simple stomach or monogastric animal.

stomach is about 100 cm to 140 cm in length and about 40 cm in diameter.

Abdominal cavity: Twenty pairs of ribs. Draw a vertical straight line between the last rib to lower abdominal.

Vertical incision down to the ground surface. Open the cavity

Stomach

Examine the stomach contents food material, moisture and gut parasites and focal ulcerations or gastritis.

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Slide 12 — Field photographs: skinning of elephant carcass team procedure

7.1 Unique Anatomical Points Critical for Necropsy

- NO PLEURAL SPACE: lungs fused to chest wall — firm adhesion is NORMAL; no thoracocentesis possible
- BIFID HEART APEX: double-pointed cardiac apex — distinctive and normal; do not record as anomaly
- ABDOMINAL TESTES: males retain testes permanently in abdomen near kidneys — NORMAL (not cryptorchidism)
- CARDIAC WEIGHT: 12–21 kg; proboscideal artery prominent in thoracic cavity
- TEMPORAL GLAND: between eye and ear; musth secretion; examine bilaterally at necropsy

7.2 Skinning Protocol

- Long median dorsal incision from mandible angle to tail base
- Reflect skin over entire lateral thoracic + lower abdominal + suspected regions
- Examine subcutaneous tissues: abscesses, haemorrhage, oedema, parasitic cysts

- Examine muscles: pallor (myopathy), haemorrhage (electrocution), atrophy
- Incise superficial lymph nodes: mandibular, parotid, prescapular, femoral

8. Digestive System & Systematic Organ Examination

8.1 Digestive System Anatomy

Elephant is MONOGASTRIC (simple-stomached) hindgut fermenter. Digestive efficiency ~40–45% of enormous daily intake (150–300 kg fresh forage). GI components: pharynx → oesophagus → stomach → small intestine → caecum (~90 L) → large intestine → rectum → anus.

Organ	Key Features	Necropsy Examination Points
Stomach	100–140 cm long; 40 cm diameter	Contents (moisture, parasites); focal ulcers/gastritis; Gongylonema in mucosa
Small Intestine	Primary digestion/absorption; very long	Intussusception; haemorrhage (salmonellosis); Quilonia/Pfenderius parasites
Caecum	~90 L; primary hindgut fermentation	Bloat; impaction; parasitic nodules
Liver	36–45 kg; multilobulated	Abscesses; fatty change; TB granulomas; Fasciola
Lymph Nodes (mesenteric)	Immune barrier for gut pathogens	Salmonella; Johne disease; parasitic

8.2 Systematic Organ Examination Checklist

System	Organs	Key Examination Points
Respiratory	Trachea, Lungs	Mucosa (haemorrhage, parasites); lung lobes (TB nodules, EEHV, consolidation)
Urogenital	Kidneys, Bladder, Uterus/Testes	Kidneys reniculated-multilobulated; testes abdominal in males (NORMAL); cystitis; endometritis
Musculoskeletal	Muscles, Bones, Joints	Haemorrhage (electrocution); osteomyelitis (captive foot disease); fractures
Reproductive	Uterus, Ovaries / Testes	Bifurcate uterus; gravidity; orchitis
Nervous	Brain, Spinal Cord	Negri bodies (rabies); haemorrhage; oedema

9. Lymph Nodes & Histopathology

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Examine every system in a systematic way
 Respiratory : Trachea, lungs
 Urogenital system : Kidney and bladder -
 Musculoskeletal system: Muscles, bones
 Reproductive systems: Uterus, etc
 Nervous system: Brain

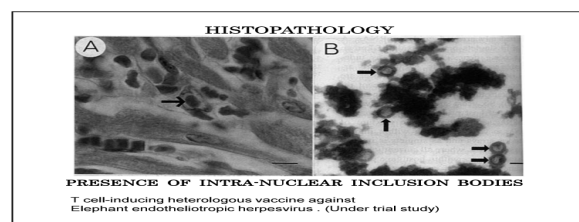
In PM examination - **Superficial lymph nodes are imp.**
 for infectious diseases i.e. Tuberculosis or EEHV.
 Locations of L nodes are largely similar to other large mammals.
 But due to thick skin difficult for external palpation.
1. Important Superficial Lymph Node Locations
 • **Mandibular Lymph Nodes:** Located at ventral border of the lower jaw, just under the skin. They are examined after incising between the mandibles to pull the tongue forward during oral cavity inspection.
 • **Parotid Lymph Nodes:** Just below the ear, near the parotid salivary gland.
 • **Superficial Cervical Lymph Nodes:** Positioned on the lateral side of the neck, slightly cranial to the shoulder.
 • **Prescapular Lymph Nodes:** Located just ahead of the shoulder joint.
 • **Femoral Lymph Nodes:** Located in an area close to head of femur.

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Slide 13 — Systematic examination list; superficial lymph node locations in elephant

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2. Deeper/Internal Nodes
 In veterinary protocols - examination of lymph nodes are imp.
Bronchial (Tracheobronchial) Lymph Nodes:
 Located at the junction where the bronchi split from the trachea. These must be sampled for mycobacterial cultures (TB).
Hepatic & Splenic Lymph Nodes:
 Located near the liver and spleen, respectively, and frequently checked for signs of systemic infection or haemorrhage.
Mesenteric lymph nodes:
 Clusters of immune cells located in the mesentery. They filter lymph fluid from the bowel and act as an immune barrier to trap bacteria, viruses, and food antigens before they enter the bloodstream.



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Slide 14 — Histopathology: EEHV intra-nuclear inclusion bodies (A & B); deep lymph nodes

9.1 Superficial Lymph Nodes

Lymph Node	Location	Key Diseases
Mandibular	Ventral border of lower jaw, under skin	TB caseous granulomas; EEHV; strangles-like
Parotid	Below ear, near parotid salivary gland	TB; systemic infection
Superficial Cervical	Lateral neck, cranial to shoulder	TB; trypanosomiasis
Prescapular	Just ahead of shoulder joint	Systemic disease; TB; lymphoma
Femoral	Near head of femur (inguinal region)	TB; pelvic infections; lymphosarcoma

9.2 Deep / Internal Lymph Nodes

Lymph Node	Location	Diagnostic Use
Bronchial (Tracheobronchial)	At carina — junction of bronchi/trachea	MANDATORY: mycobacterial culture (TB); EEHV PCR
Hepatic & Splenic	Near liver and spleen	Systemic infection; TB hepatic involvement
Mesenteric	Mesentery of small/large intestine	Intestinal infections; enteritis; parasites

9.3 EEHV — Histopathological Diagnosis

Elephant Endotheliotropic Herpesvirus (EEHV) is the leading infectious cause of death in captive elephant calves worldwide. Definitive diagnosis: INTRA-NUCLEAR INCLUSION BODIES (Cowdry Type A) in vascular endothelial cells of heart, tongue, trunk skin, liver, spleen.

- Gross: petechial/ecchymotic haemorrhages on serosal surfaces; haemopericardium; oedematous head/neck
- Histopathology: swollen endothelial cells with large eosinophilic intranuclear inclusions surrounded by clear halo
- PCR confirmation: trunk wash, blood, tissue samples — must be FRESH (unfixed)
- 7 EEHV species (EEHV1–7); EEHV1A and EEHV4 most commonly fatal; median survival ~36 hrs
- Vaccine: T cell-inducing heterologous vaccines currently under trial

EEHV SAMPLING: trunk wash (fresh) + blood (EDTA) + heart/tongue/trunk skin (fresh frozen + formalin) + lymph nodes (fresh) → send to IVRI/national EEHV reference laboratory. Time critical — act within 36 hrs of signs.

10. Differential Diagnosis & Case Studies

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DIFFERENTIAL DIAGNOSIS	
Diseases with sudden onset and rapid death	
Haemorrhagic septicaemia	Extensive haemorrhagic areas
Salmonellosis	Watery diarrhoea with or without blood.
Enterotoxaemia	Diarrhoea may lead to dehydration and deaths.
Encephalomyocarditis	Respiratory signs may be observed, some times.

Rabies in semi-captive elephant at Panna Tiger Reserve



Work body condition

- Initially, off fed, restlessness and drooling of saliva
- 3rd day, restricted jaw movement and protrusion of tongue
- 4th day, hyperaemic conjunctiva
- 5th day, hydrophobia, loud but changed trumpeting
- 6th day, Lateral recumbency and death
- Proper disposal of carcass



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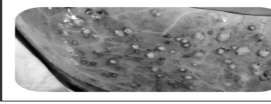
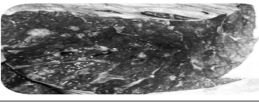
Slide 15 — Differential diagnosis: sudden death diseases; Rabies case at Panna Tiger Reserve

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Lesions of Tuberculosis in free range Asian elephant



Photo: Dr. Parag Nigam

Difficult to advise



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Slide 16 — TB lesions in free-range Asian elephant (Photo: Dr. Parag Nigam); decomposed PM

10.1 Differential Diagnosis — Sudden Onset Rapid Death			
Disease	Agent	Key PM Finding	Priority Sample
Haemorrhagic Septicaemia	Pasteurella multocida B/E	Extensive haemorrhages; pulmonary oedema; splenomegaly	Blood anaerobic; lung; spleen for culture + PCR
Salmonellosis	Salmonella spp.	Haemorrhagic enteritis; watery diarrhoea ± blood; mesenteric LN haemorrhage	Intestinal content; LN; blood culture; faeces PCR
Enterotoxaemia	Clostridium perfringens C/D	Haemorrhagic enteritis; gas in gut; rapid autolysis	Intestinal content frozen for toxin ELISA; PCR
Encephalomyocarditis	EMC Virus	Pale foci in myocardium; encephalitis; respiratory signs	Heart + brain fresh + formalin; blood serology + PCR

Rabies	Rabies Lyssavirus	Negri bodies in brain stem; non-suppurative encephalitis	Brain stem fresh (FA test); brain formalin (histology)
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10.2 Rabies Case — Panna Tiger Reserve (Dr. A B Shrivastava)

Day	Clinical Signs
Day 1	Off feed; restlessness; drooling of saliva (hypersalivation)
Day 3	Restricted jaw movement; protrusion of tongue (paresis)
Day 4	Hyperaemic conjunctiva (autonomic involvement)
Day 5	Hydrophobia; loud but changed, distorted trumpeting (laryngeal paralysis)
Day 6	Lateral recumbency and death (ascending paralysis complete)
Post-mortem	Proper disposal — deep burial or incineration; biohazardous; PEP for all contacts

10.3 TB Lesions in Free-Range Asian Elephant

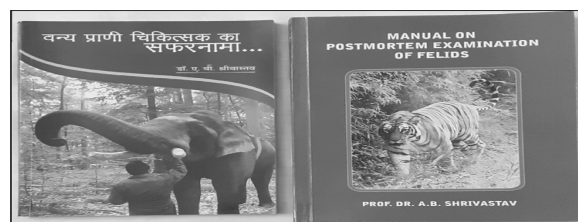
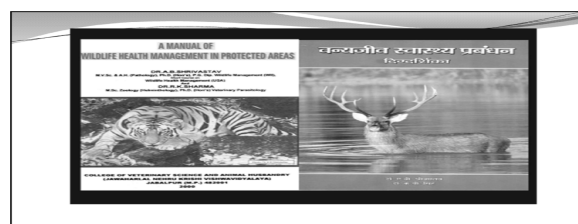
- Causative agent: *Mycobacterium tuberculosis* (HUMAN strain) — fully zoonotic
- Gross: grey-white caseous nodular granulomas in lungs; miliary nodules on pleura/peritoneum; caseous/calcified lymph nodes
- Collect: tracheobronchial LN (priority); lung nodules; trunk wash → culture (8–12 weeks) + PCR
- ALL personnel: N95 mask + double gloves + goggles; mandatory post-exposure TB screening

10.4 HCN Poisoning — Captive Elephants (Zoo Print 2010)

- Cause: Sorghum (Jowar) fed to animals unaccustomed to it — cyanogenic glycosides released
- Signs: restlessness, diarrhoea, subnormal temperature 97–98°F
- Treatment: Sodium thiosulphate 50 gm oral; dextrose saline + Diulin rectally
- PM: cherry-red blood/tissues; bitter almonds smell; congested viscera

11. Key References & Recommended Reading

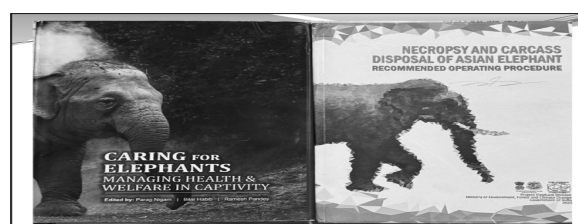
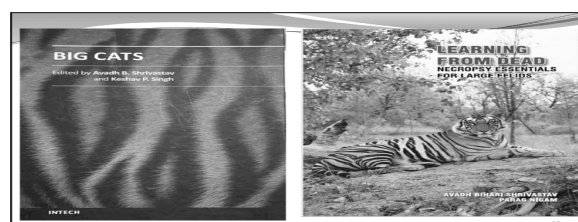
06-06-2026



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Slide 17 — Published works of Prof. Dr. A B Shrivastava (Vol. 1)

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Slide 18 — Published works: Learning from Dead; Caring for Elephants; Necropsy SOP

11.1 Works by Prof. Dr. A B Shrivastava

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11.2 Peer-Reviewed Literature

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Prof. Dr. A B Shrivastava, NDVSU Jabalpur | Presentation: 06-06-2026 | Training Program: Mortality Investigation of Asian Elephant, Raigarh, Chhattisgarh | Notes elaborated with peer-reviewed literature